Conference Program

War Resilient Agrifood Biotech & AI

Satellite Event of the European Climate Pact

Format: 100% Remote | Asynchronous Video | Interactive Digital Posters

Introduction

Wars in 2026—escalations in the Middle East (Gaza, Lebanon, Iran-Israel), Sudan, Ukraine, and others—have created war driven food crises. To overcome these bottlenecks and to discuss how climate-adaptive crops can empower disrupted supply chains, we invite the Startups, the new researchers and practitioners, students and talented policymakers to discuss and share insights and prospective how low-tech biotech for smallholders in unstable regions might help. The conference is Digital Poster and Asynchronous Video based, to support also the conference attendance in affected regions.

Conference Objectives

- Highlight past achievements, recent advancements and future trends in AI based biotechnologies and their transformative potential for achieving efficient, inclusive, resilient and sustainable agrifood systems.
- Explore how biotechnologies, including low-tech tools and advanced innovations, such as genome editing and synthetic biology, small-scale producers, processors, traders and retailors of climate-adaptive crops can empower disrupted supply chains, low-tech biotech for smallholders in unstable regions
- Share good practices in integrating digital technologies, to advance research, improve access, and scale biotechnological innovations.

Expected outcomes: The expected outcomes of the conference include fostering knowledge exchange on a wide range of biotechnological options for agrifood system transformation and increased awareness of the potential of biotechnologies with a Call for Action for achieving efficient, inclusive, resilient and sustainable agrifood systems.

The consolidated summaries of fully remote prerecorded plenaries and technical sessions are expected to inform the update of the EU statement call for evidence on biotechnology after the conference.

Target audience: The target audience for the conference includes representatives of Startups, PhD students, governments, international and inter-governmental organizations, academia and research institutions, civil society organizations, private sector (including philanthropic foundations) and producer organizations.

Format: The conference will consist of technical sessions on thematic topics, prerecorded plenaries focusing on Agrifood system sub-themes, Rapid Innovation Reports, "Tech-Brief" Session for startups and PhDs and the 5-Minute Pitch" Session (Rapid Research & Startup Demos). The individual sessions will be organized by adopting innovative and interactive formats.

There are 9 main themes for the conference. Four prerecorded plenaries will be dedicated to each main theme. Note that, the titles of the sessions are not finalized and those provided below are only indicative.

- The "DOI Guarantee": Every accepted poster is assigned a unique DOI, ensuring young researchers get immediate, citable credit for their work.
- Submission Categories:
    - Posters: Digital A0 landscape format. Ideal for ongoing PhD research and startup tech-previews.
    - Video Presentations: 10–15 minute MP4 files. In English/French/ Spanish. Best for young professors and innovators to showcase "AI in Action."
- The call emphasizes " Tech-to-Market" and "AI-Driven Sustainability" focused on the startup ecosystem.
- Asynchronous "Degrees of Freedom": Participants can engage with the video gallery at their own pace, which is a "warm welcome" for researchers across different time zones.
- Q&A forum with live annotations [asynchronous comments | voting | peer discussion | best poster]
- This is a Satellite event of European Climate Pact
- A distributed volunteer editorial team of 120 splendid researchers and practitioners from Universities and startups from Global South, will Review, revise, and deal with the technical blithers of the event. This splendid team of volunteer will add subtitles in the video for non-native speakers and will share with the participants the Digital Certificate of Participation with a block chain verification of recognition path.
- The conference presentations will be evaluated by 3 (three) independent Experts.
- Post Conference Phase: Publication of the Abstract Book and Conference Proceedings Book with ISBN and ISSN and the building of the VOD pipeline.
- The repository of the conference Proceedings is Zenodo
- Google Scholar Indexed
- The conference follows EU Metadata Policies and Standards
- 12 Scholarships for participants from Global South and LIC
- **Conference Deadlines:**
    - **Abstract Submission: May 25, 2026.**
    - **Notification: June 15, 2026.**
    - **Conference Launch: July 10, 2026.**
    - **Post Conference: October 10, 2026**

RESEARCHERS & EXPERTS SESSION

- Facing the challenges of climate change: Adaptation in the crop and forestry sectors
- Improving productivity through enhanced resource use efficiency
- Social and economic impacts of agricultural biotechnologies for smallholders: Taking stock of the evidence and prioritizing future assessments

Facing the challenges of climate change: Adaptation in the crop and forestry sectors	Improving productivity through enhanced resource use efficiency	Social and economic impacts of agricultural biotechnologies for smallholders: Taking stock of the evidence and prioritizing future assessments
Developing rice varieties with enhanced adaptation to lowland farming systems	Resource use efficiency in crops: “Green super rice” to increase water and nitrogen use efficiency of rice	Evidence-based policy-making: The role of impact assessment studies and their implications for agricultural biotechnologies
Harnessing agricultural biotechnology for resilience to climate change	Resource use efficiency in vegetables: Application of molecular breeding to underutilised crop for low-input agriculture	Country experience in the use of evidence in policy-making on agricultural biotechnologies
Molecular breeding in legumes for resource-poor farmers	Resource use efficiency in fish: Application of biotechnology in genetic improvement in aquaculture	Factors that determine whether biotechnologies can have positive impacts on the livelihoods of smallholders
Genomic approaches for dissecting fitness traits in forest tree landscapes	Resource use efficiency in livestock: Bridging the biotechnology-livestock productivity gap	
	Resource use efficiency in forestry: Utilization of tree genetic resources	
Closing Remarks		

STUDENTS AND STURTUPS INTERACTIVE SESSION

- Climate change: Adaptation in the livestock and fishery sectors
- Post production value addition and food safety (2 sub-sessions)
- Public policies, strategies and regulations on agricultural biotechnologies

Climate change: Adaptation in the livestock and fishery sectors	Post production value addition and food safety	Public policies, strategies and regulations on agricultural biotechnologies
Fish breeding for future environments under climate change	Use of feed additives generated through fermentation technologies for livestock feed	The role of intellectual property rights in enabling or impeding the application of agricultural biotechnologies, with special reference to developing country agriculture
Development of diagnostic tools and vaccines for aquatic animals	Use of biotechnologies to increase the storability and shelf life of fruit	The role of intellectual property rights in enabling or impeding the application of agricultural biotechnologies, with special reference to developing country agriculture
Development of diagnostic tools and vaccines for aquatic animals	Applying agricultural biotechnology tools and capabilities to enhance food security and nutrition from local food crops to stimulate sustainable income opportunities for small holder farmers to reduce poverty	Regulation and intellectual property of agricultural biotechnologies: Perspectives from the private sector
Biotechnologies for animal breeding and coping with climate change	Using biotechnologies to ensure the safety of food	Regulation and intellectual property of agricultural biotechnologies: Perspectives from the civil society
Use of biotechnologies to improve feed quantity and quality: Adaptation to the changing climate from the animal nutrition perspective	Aflasafe: a case study for aflatoxin reduction in crops	The challenges of developing national policies and regulations for agricultural biotechnologies: Reflections from cumulative experience
A landscape genomics approach in unravelling adaptive genetic diversity in goat	Diagnostic tools to detect pathogens causing tuberculosis in cattle and prevent their transmission through dairy products to humans	How to ensure that policies, strategies and regulations on agricultural biotechnologies act to benefit smallholders

‘Tech-to-Market/ AI-Driven Sustainability’ SESSIONS DAY 3

Sustainable Agriculture and Natural Resources Climate-smart agriculture and sustainability Soil, water and biodiversity management Sustainable land use and agroecology Ecosystem services and resilience	Crop Science and Horticulture Crop production systems and innovation Fruit growing and protected cultivation Plant physiology and stress adaptation Sustainable plant protection	Animal Science and Veterinary Systems Animal health and welfare Livestock systems and pastoralism Veterinary public health One Health approaches
Food Science, Safety and Nutrition Food quality and safety Food processing and innovation Nutrition and public health Functional and traditional foods	Biotechnology and Life Sciences Agricultural biotechnology Molecular tools and genomics Microbiome and plant health Emerging bio-based innovations	Environment and Climate Change Environmental health and pollution Climate change impacts and adaptation Water and ecosystem management Biodiversity conservation
Rural Development and Agri-food Systems Rural economies and livelihoods Value chains and agribusiness Agrotourism and local development Socio-economic transformation	Policy, Governance and EU Integration Agricultural and food policies EU standards and integration Sustainability transitions Evidence-based policymaking	Digitalization and Innovation in Agriculture Precision agriculture and data tools Smart farming technologies Digital food traceability Innovation ecosystems
Closing Remarks		

Agenda may be subject to change